

MID SEM. EXAMINATION, Sep.-2025

Course Code: ECECC305/EIECC305/ICECC305/VTECC305

Course Title: Digital Circuits and Systems

Time: 01:30 Hrs.

Max. Marks: 20

Note: Attempt all questions. All questions carry equal marks. Missing data/information if any must be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
1	(a) If $(10W1Z)_2 \times (15)_{10} = (Y01011001)_2$ then find the value of W, Y and Z. (b) Solve the following function using K-map and provide the answer in POS form. $F = \sum m(0, 1, 3, 4, 6, 8, 9, 10) + d(5, 7, 12, 13, 14, 15)$.	2+2	CO1
2	(a) Write the complete table to simplify the function $F(w,x,y,z) = \sum (0,1,2,8,10,11,14,15)$ using the tabulation method/Quine-Mc-Cluskey Method. (b) Draw its prime implicant chart and give the simplified expression of F.	2+2	CO1
3	(a) Write the truth table for all the outputs of a combinational circuit that generates 1's complement of BCD code. (b) In the circuit given below, find the expression for X	2+2	CO2
4	(a) Give the simplified expressions for all the possible outputs related to a circuit that performs the addition cum subtraction function given by the following expression: $f = x + y - z$ Where x, y, and z are three single bit inputs. (b) Design the following function using a complemented 3X8 decoder and basic gates. Realize it using minimum hardware. $f = \sum (0,1,4,5,6)$	2+2	CO2
5	Convert a negative edge triggered D flip-flop into negative edge triggered JK flip-flop. (a) Write the required table for the conversion. (b) Find the required expressions and draw the complete circuit.	2+2	CO3